

Rural Road Access and Livestock Production in Ethiopia

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Abstract

Although livestock plays a vital role in rural livelihoods across Sub-Saharan Africa, productivity in the region remains low due to suboptimal management practices, low-yielding breeds, and recurrent disease outbreaks. This paper examines whether improved rural connectivity can improve the livestock production system using Ethiopia's large-scale rollout of the Universal Rural Roads Access Program as a quasi-experimental setting. We merge road network data with multiple rounds of nationally representative household surveys collected throughout the road expansion period. To address endogeneity in road placement, we implement an instrumental variables strategy that exploits variation in topography and pre-existing administrative budget allocations to identify the causal impact of road access on herders. We find that improved road access leads to significant gains in livestock production, including greater adoption of improved cattle breeds, the increased use of veterinary services and purchased feed, and a reduced reliance on open grazing. Road access also significantly lowers the likelihood of cattle mortality. The findings highlight the potential to improve the livestock systems in the region through improved road connectivity.

Keywords: Rural roads, livestock production, agricultural development, instrumental variables

JEL Classification: R42, Q12, Q16, O18

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1. Introduction

Livestock production supports livelihoods in Sub-Saharan Africa in several ways. It provides nutritious food (Hirvonen and Hoddinott, 2017; Fadare et al., 2024), generates income for consumption and investment (Herrero et al., 2013; Do et al., 2022), and provides resilience against shocks (Acosta et al., 2021). However, poor husbandry practices, reliance on low-yield breeds, and frequent disease outbreaks constrain its productivity in the region (Ahmed et al., 2018, 2022; Rufael et al., 2008; Tschopp et al., 2022).

Better road connectivity could reduce some of these constraints. For instance, improved road access may improve animal health by increasing access to better feed and veterinary services (Ochieng and Hobbs, 2016). Improved roads could also encourage investment in high-yielding breeds and related technology by making essential inputs accessible and selling perishable products easier (Abay and Jensen, 2020). However, road access may also negatively affect the agriculture sector as it creates more attractive non-agricultural employment opportunities, drawing labor away from agricultural activities (Asher and Novosad, 2020). As a result, the net impact of rural roads on livestock production is theoretically ambiguous and remains an empirical question.

This paper examines the impact of road access on livestock production using Ethiopia’s Universal Rural Roads Access Program (URRAP) as a case study. Under the URRAP, the Ethiopian government constructed more than 62,000 kilometers of all-weather roads between 2011 and 2016 (Gebresilasse, 2023; Kebede, 2022). The program increased the share of villages with road access from 42 percent to over 76 percent. We leverage this expansion to examine its effects on smallholder livestock production, with a particular focus on cattle-keeping communities.

Identifying the causal effects of road access is challenging due to the endogeneity of road placement decisions. For example, policy makers might prefer to construct roads in areas expected to provide significant socioeconomic or political benefits (Perra, 2022; Burgess et al., 2015; Jedwab and Moradi, 2016). To address this, we employ an instrumental variables approach that exploits exogenous variation in natural factors such as the slope of terrain, existing land use, baseline proximity to roads, and regional budget allocation decisions made before the implementation of the expansion program. These factors are plausibly exogenous because they are determined by longstanding geographic and historical factors rather than contemporary socioeconomic characteristics.

For the analysis, we combine a geospatial dataset on road infrastructure with a nationally representative livestock rearing household survey collected annually from 2011 to 2016. Our results

show that road access is significantly associated with substantial improvements in livestock production. Specifically, road access is associated with increased adoption of exotic or hybrid cattle breeds, higher vaccination rates, greater use of improved feeding practices, and lower livestock mortality rates. More precisely, we find that road access is associated with a 5.7 percentage point reduction in the use of grazing as a source of feed and a 4.7 percentage point increase in the likelihood of feed purchase. Vaccination rates also rise by 8.4 percentage points in connected areas. We also find that the presence of a road in the village is associated with a 5.3 percentage point reduction in the probability of cattle deaths. These results suggest that improved road connectivity can be considered a potential strategy to improve the performance of the livestock sub-sector in rural areas.

The study contributes to the literature that investigates the socioeconomic impacts of rural connectivity. Existing related studies primarily focus on their effects on household income and consumption dynamics ([Dercon et al., 2009](#); [Jacoby and Minten, 2009](#); [Stifel and Minten, 2017](#); [Asher and Novosad, 2020](#)), crop production and farm technology use ([Aggarwal, 2018](#); [Gebresilasse, 2023](#); [Shamdasani, 2021](#); [Damania et al., 2017](#)), market integration ([Shively and Thapa, 2017](#); [Kebede, 2024](#); [Jones and Salazar, 2021](#)), education ([Adukia et al., 2020](#)), health ([Aggarwal, 2021](#)), and labor mobility ([Asher and Novosad, 2020](#); [Dappe and Lebrand, 2024](#)), with relatively little attention to livestock production.

Our study extends this literature by analyzing rural road investments with a specific focus on the livestock sector. More precisely, we contribute to the literature by examining how improved road infrastructure affects livestock management practices, including the adoption of improved breeds, the utilization of veterinary services, feeding strategies, and the reduction of livestock mortality in cattle-keeping households.

There are pertinent reasons to focus on Ethiopia. Ethiopia has the largest cattle population in Africa, with more than 70 million heads ([FAO, 2025](#)). However, its productivity falls below expectations. [Livestock Systems Innovation Lab \(2021\)](#) identified, among others, poor feeding practices, widespread disease, the low genetic productivity of indigenous breeds, and limited market access as the main reasons for its low productivity. For example, [Ethiopian Statistics Service \(2023\)](#) showed a low level of adoption of commercial and improved feed in the country’s livestock production system. The report lists open grazing, green fodder, crop residues, and hay as the main sources of cattle feed in the country. Animal health is another challenge in the country. Relatedly, [Tschopp et al. \(2009\)](#); [Kaba \(2022\)](#); [Rasmussen et al. \(2024\)](#) have attributed the high mortality and reduced

productivity of the livestock sub-sector to diseases such as foot-and-mouth disease, lumpy skin disease, and brucellosis. Access to effective extension services could address some of these issues. However, fewer than one percent of farmers in the country have access to livestock-related extension services ([Ethiopian Statistics Service, 2023](#)). Livestock keepers in the country also face significant challenges in accessing markets and other services. For example, [Leta and Mesele \(2014\)](#) shows that only 46% of cattle in the country are reared within a 10 km radius of major livestock market centers and all-weather roads. Such isolation is expected to restrict farmers’ market participation. Relatedly, [Abay and Jensen \(2020\)](#) argued that only 38% of livestock keepers in the country raise animals primarily for market purposes. Hence, this study empirically examines whether access to rural road infrastructure helps alleviate some of the challenges associated with livestock production.

The remainder of this paper is organized as follows: Section 2 describes the data sources and variable definitions. Section 3 outlines our methodology, including the construction of instrumental variables. Section 4 presents the results, discusses their implications, and presents robustness checks. Section 5 concludes.

2. Data Sources and Types

2.1. Annual Livestock Sample Survey

We use the annual livestock sample survey (ALSS) from the Central Statistical Service (CSS) of Ethiopia to construct our outcome and control variables ([Ethiopian Statistical Service, 2011](#)). ALSS annually targets rural agricultural households across all regions, excluding those with non-sedentary lifestyles. It follows a two-stage stratified cluster sampling design. In the first stage, enumeration areas (EAs) are selected from the Population and Housing Census cartographic frame using a probability proportional to size yielding a sample of 2,273 EAs.¹ In the second stage, 30 agricultural households are systematically selected from each EA using an updated household list available at the time of the survey. We use repeated cross-sectional data collected between 2011 and 2016. During these years, the CSS consistently used the same EAs each year, while resampling households each year. Given that CSS resamples a fixed number of households (30) from the same EA each year, the samples in each wave are expected to be comparable. To assess their consistency

¹Ethiopia’s administrative hierarchy comprises four levels: regions, zones, woredas, and kebeles. Kebele represents the lowest administrative unit in the country. EA under ALSS generally encompasses a geographic area equal to or smaller than a rural kebele and typically includes 150 to 200 households ([Ethiopian Statistical Service, 2011](#); [Ethiopian Statistics Service, 2023](#)).

across survey rounds, we present the descriptive statistics for key household characteristics, such as age, sex, education of the household head, family size, and production orientation over time in Table A1.

Our study focuses on cattle-keeping households to avoid aggregation issues that arise from combining multiple species with distinct production characteristics and responses to external factors. Additionally, improved production systems are predominantly concentrated within the cattle subgroup, making it the most relevant for examining the impact of infrastructure on livestock productivity and management.

We construct three groups of outcome variables from ALSS. The first group involves household cattle ownership and breed composition. We use two indicators for this: the total number of cattle, which measures herd size, and the proportion of hybrid or exotic breeds, calculated as the fraction of hybrid/exotic cattle relative to the total herd. The second group focuses on cattle health and is represented by two variables. The first is a binary indicator for whether a household experienced at least one cattle death. The second is the proportion of vaccinated cattle, calculated as the number of vaccinated cattle divided by the total herd size. The last group focuses on feeding practices with two variables. The first is a binary indicator for whether the household purchases feed (1 for purchase, 0 otherwise). The second variable reflects the share of grazing in total feed, expressed as a fraction.

2.2. Road Network

We use a road network shapefile accessed from the Ethiopian Road Authority (ERA) to construct our key explanatory variable. The shapefile provides a detailed map of all roads in the country and geographic data for spatial analysis. It also includes metadata on the construction history of each road segment, which allows us to distinguish kebeles that gained road during our study period from those that did not. Figure 1 shows the expansion of the rural road network under URRAP.²

For the analysis, we excluded kebeles with pre-existing road infrastructure. Hence, the analysis focuses solely on kebeles that either received a new road or remained untreated during the study period. Among the 2,085 kebeles included in the ALSS, 371 never received a road, 872 received a new road during the study period, and 842 had pre-existing roads at baseline. Descriptive statistics in Table A2 summarize key variables focusing on kebeles that received new roads or remained

²URRAP initially aimed to cover all regions but excluded the Afar and Somali regions (Kebede, 2024). After removing the two regions, we were left with 97% of our observations.

untreated. ALSS interviewed 128,497 cattle-keeping households from these kebeles during the study period.

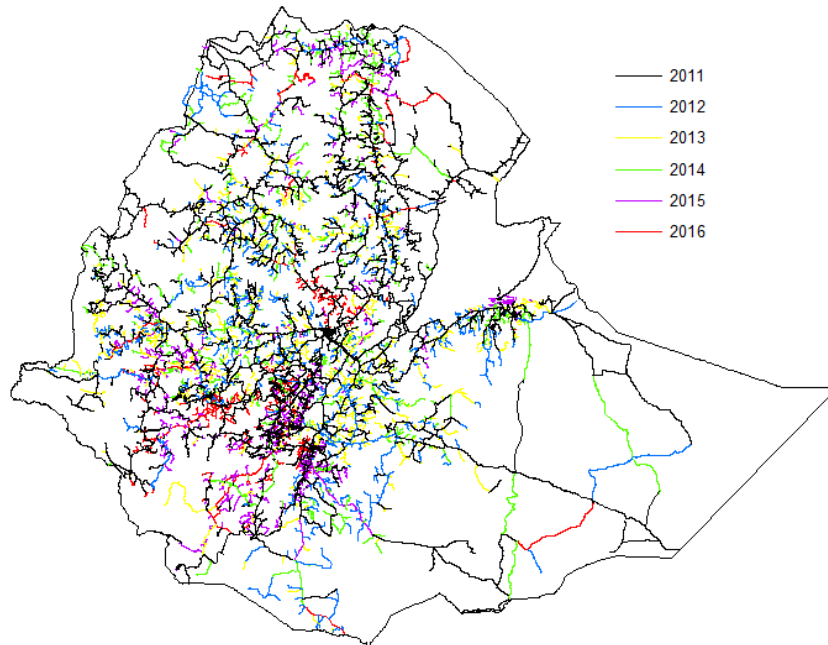


Figure 1: Map of the annual road network, 2011 - 2016.

Note: The figure shows the road segments constructed each year during the study period, with colors corresponding to each year as indicated in the legend.

2.3. Additional Data Sources

We utilized additional datasets to control for environmental and agricultural factors that influence road placement and livestock production outcomes. We use the Standardized Precipitation-Evapotranspiration Index (SPEI) to account for water availability. The SPEI offers a more comprehensive view of water balance by incorporating both precipitation and potential evapotranspiration. We employ a high-resolution SPEI dataset ($0.05^\circ \times 0.05^\circ$, ≈ 5.5 km), developed by [Peng et al. \(2020\)](#). The index is standardized to have a mean of zero. Values near zero indicate normal water availability. Positive SPEI indicates wetter-than-average conditions, and negative values indicate drought.³

We also use the Global Agro-Ecological Zones (GAEZ v4) dataset to account for variations in

³We use a one-year lag value since the availability of feeds (such as crop residues) in the current year is mostly influenced by rainfall conditions in the previous year.

productivity potential that could influence road infrastructure development and livestock production (FAO and IIASA, 2021). GAEZ provides suitability indices for various crops based on factors such as topography, soil characteristics, and input assumptions. The GAEZ dataset has a 30 arc-second (approximately 0.9 x 0.9 km) resolution. We use the baseline data from 1961 to 2010 to address potential endogeneity related to agricultural potential and road access.⁴ We use the kebele shapefile to merge geospatial datasets with the annual livestock sample survey by calculating zonal mean values for kebeles.

3. Estimation Strategy

Estimating the impacts of road infrastructure on livestock production is challenging due to the endogeneity associated with the non-random nature of road placement. If the road placement decision was random, its impact could be estimated using the following simple regression model:

$$Y_{ivt} = \beta_0 + \beta_1 \text{Road}_{vt} + \beta_2 X_{ivt} + \epsilon_{it} \quad (1)$$

where Y_{ivt} represents livestock production outcomes for farmer i in kebele v at time t . Road_{vt} is a binary indicator of road presence, X_{ivt} is a vector of control variables, and ϵ_{ivt} is the error term. β_1 would provide an unbiased estimate of the impact of roads on livestock production under the random placement assumption.

However, policymakers usually allocate roads based on their expected economic gains (Coşar and Demir, 2016; Burgess et al., 2015; Asher and Novosad, 2020; Jedwab and Moradi, 2016). Political incentives also play a role in road placement decisions (Perra, 2022; Blimpo et al., 2013). One mitigating factor in the Ethiopian context is that road improvement plans were devised as part of a five-year strategic national development program (Shiferaw et al., 2015; Perra et al., 2024; Fiorini et al., 2021). Specific to the URRAP, the level of such bias is minimal, as its primary objective was to connect all rural kebeles to baseline roads.⁵ However, even within this framework, economic potential, geographic, and budgetary constraints likely influenced the sequencing of road construction.

If roads were allocated based on eligibility criteria, such as population thresholds, these cut-off

⁴For more information on the GAEZ methodology, visit <http://www.fao.org/nr/gaez/en/>.

⁵The Ethiopian Road Authority states URRAP’s mission as: “To connect all kebeles by all-weather road and provide year-round access, ensuring that all road infrastructure meets the needs of rural communities and remains affordable to build and maintain” (Ethiopian Roads Authority, 2012).

points would serve as a natural experiment for evaluating the impacts (Asher and Novosad, 2020). However, the Ethiopian Road Authority does not publicly provide the eligibility criteria. Hence, our empirical strategy employs instrumental variables (IV) derived from exogenous factors that influence road placement decisions. More precisely, we follow Kebede (2022), and Gebresilashe (2023) and construct our instrument using terrain slope, land use types, distance to baseline road, and the regional road expansion budget set before the commencement of the program. The instrument is based on the idea that road allocation decisions are influenced by two factors: (1) construction costs, which are shaped by geographic features and proximity to the baseline road, and (2) the sequence of construction, which is driven by annual budget constraints. Accordingly, we calculate the least-cost path from each rural kebele centroid to the baseline road network using algorithms from civil engineering and network theory. We create our IV by allocating roads sequentially to the rural kebeles until the regional road expansion budget is exhausted.

We follow the following steps to construct our IV. We first develop a cost raster using publicly available Digital Elevation Model (DEM) (Jarvis et al., 2008) and land use data from (Bartholomé and Belward, 2005). The DEM is used to estimate terrain slope, while different land use types are assigned weights based on their construction difficulty. Then, we use the cost-distance analysis tool in ArcGIS to calculate the least-cost paths from each village centroid to the nearest point on the baseline road network. Lastly, we apply a cost-minimization strategy that aligns with regional budget constraints set by national road authorities before the start of URRAP. Hence, we prioritize constructing less expensive roads first under a fixed budget, and roads are sequentially allocated from nearest to farthest until the cumulative length reaches the annual budget limit for each region. Details on these budgets are provided in Table A3.

After constructing our IV, we apply a two-stage instrumental variable approach. In the first stage, we regress the endogenous variable, road placement, on the IV and other controls to obtain predicted values for road placement as shown in equation 2:

$$Road_{ivt} = \pi_0 + \pi_1 IV_{ivt} + \pi_2 X_{ivt} + \delta_t \left(\text{Distance}_v^{\text{base}} \times D_t \right) + \mu_i + \lambda_t + \gamma_{rt} + \epsilon_{ivt} \quad (2)$$

where, $Road_{ivt}$ is a binary indicator denoting the presence of a road in kebele v at time t ; IV_{it} represents the predicted road placement decision (1 if the kebele is identified to receive a road, 0 otherwise), while ϵ_{it} represents the error term. X_{ivt} includes control variables that capture factors potentially confounding the relationship between road infrastructure and livestock production. We

control for demographic, socioeconomic, geographic, and environmental variables. We control for the age and gender of the household head and the size of the household. We also include agricultural suitability variables from the GAEZ dataset to control for agricultural suitability, which may simultaneously influence road placement and livestock production as high crop production areas may be more likely to receive roads first and can produce more livestock feed. We also control for climate variability using a one-year lagged SPEI. To control for population density, we include kebele-level population data from the 2007 Ethiopian census. Densely populated areas are more likely to receive roads due to higher demand and political considerations. Larger populations may also drive increased demand for livestock products or provide additional labor. By using pre-expansion census data, we mitigate concerns about population shifts caused by improved road access. We include district fixed effects (μ_i) to address unobserved heterogeneity across regions, such as differences in governance, agroecological conditions, and cultural practices, which could influence both road placement and livestock outcomes. It is also important to highlight factors such as landscapes, land use, and proximity to existing roads, although not directly related to livestock production, that can still indirectly affect livestock outcomes. For instance, challenging terrain may limit the availability of grazing land, and river crossings can hinder the transportation of livestock or enhance water access for livestock. The inclusion of district fixed effects helps to account for these potential indirect effects. Furthermore, we include interactions between the distance to baseline roads and the survey year ($\text{Distance}_v^{\text{base}} \times D_t$) to address the effects of remoteness. It is expected to account for differences between kebeles that are close to the baseline roads and those that are far. We also included region-by-year fixed effects (γ_{rt}) to absorb region-specific time-varying unobserved factors that affect road placement decisions.

The predicted values of road placement, $\hat{Road}_{it}$, are derived from the first-stage regression (2). These predicted values are then used to estimate the impact of road infrastructure on livestock production outcomes, Y_{it} , in the second stage:

$$Y_{it} = \alpha + \beta \hat{Road}_{it} + \gamma X_{it} + \delta_t \left(\text{Distance}_v^{\text{base}} \times D_t \right) + \mu_i + \lambda_t + \gamma_{rt} + \epsilon_{it}, \quad (3)$$

The coefficient β denotes the Local Average Treatment Effect (LATE), the impact of road infrastructure on livestock production for producers whose access to the road is altered by the instrument. Provided that the instrument strongly predicts road construction, influences production exclusively through changes in road access, and modifies access uniformly among relevant producers,

β identifies the effect of improved road access on the subset of affected producers. Accordingly, we assess the validity of our instrumental variable by testing its relevance with a first-stage regression. Table 1 shows that the instrument is strongly correlated with road access.

We also conduct an additional test to assess whether baseline livestock production potential influenced road allocation decisions. Accordingly, we examine the correlation between the year a kebele received a road and outcomes measured at the baseline (such as the number of exotic and hybrid cattle or cattle mortality rates). Results in Tables A4 and A5 show no significant relationship between baseline outcome variables and the timing of road placement. Hence, we do not find clear evidence of systematic differences in outcome variables between kebeles that received roads earlier and those that received them later.

4. Results and Discussions

4.1. Main Results

In this section, we present the findings of the instrumental variables approach. As shown in Table 1, road access is associated with a 0.7 cattle decrease in the total number of cattle owned. Farmers in kebeles that gained access to a road are also 1.8 percentage points more likely to own exotic or hybrid cattle.⁶ This shift suggests that improved road access encourages farmers to prioritize high-value breeds over larger herd sizes.⁷

Numerous factors linked to improved road infrastructure can explain the increase in the share of exotic and hybrid cattle due to improvement in road connectivity. For instance, high-yielding cattle breeds that produce more milk and meat are easier to obtain when road accessibility is improved. Road also makes vital inputs and services relevant for commercial farming, such as adequate veterinary support and better feeding techniques, more accessible. Roads also make it easier to sell livestock products, especially perishable commodities like milk, at higher-value urban

⁶A reduction in total herd size can mechanically increase the proportion of exotic cattle, even if their absolute number remains unchanged. Hence, the observed change in cattle breed proportion may result from either an increase in exotic cattle or a decrease in indigenous cattle. To validate the proportional findings, we use the absolute numbers of local and hybrid, or exotic cattle. The results show that road access reduces the number of local breed cattle by 0.557 (s.e. = 0.137, $p < 0.01$) and increases the number of hybrid or exotic cattle by 0.090 (s.e. = 0.012, $p < 0.01$). These findings indicate that the increase in the share of exotic and hybrid cattle reflects both genuine adoption and a reduction in indigenous herds, rather than a purely mechanical effect.

⁷There is a meaningful difference in the performance of local and exotic cattle breeds. For example, the average beef yield of local breeds in Ethiopia is approximately 110 kilograms per head, compared to the global average of 212 kilograms (Abebe et al., 2022). However, local breeds have the important advantages of being well adapted to the local ecology and exhibiting greater disease resistance.

and regional markets.

Table 1: Impact of Road on Livestock Size and Types

VARIABLES	Cattle Owned	Share of Exotic and Hybrid Cattle
Road	-0.704*** (0.196)	0.018*** (0.003)
Controls	Yes	Yes
Observations	128,497	128,497
<i>First-stage regression summary</i>		
Coefficient on instrument		0.331***
Standard error		(0.004)
Partial R-squared		0.492
F-statistic (1, 127,974)		7027.450***

Note: The table reports estimates from a two-stage instrumental variable regression analyzing the impact of road exposure on livestock outcomes. The dependent variables are total cattle owned and the share of exotic and hybrid cattle. The instrument is allocated road length. First-stage statistics confirm instrument relevance. All models include household-level controls (age and gender of household head, family size), village-level agro-climatic controls (crop suitability and SPEI), and fixed effects for time, district, and remoteness. The mean values of the control variables are presented in ???. This table also summarizes key model statistics from the first-stage regression used to assess instrument relevance. Robust standard errors are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Additionally, our findings suggest that the improved road network facilitates the adoption of improved feeding practices. As shown in Table 2, our analysis reveals that the presence of roads is associated with a significant shift away from traditional grazing-based feeding methods. We find that the presence of a road in a village is correlated with a 5.7 percentage point reduction in the share of grazing in total feed. Concurrently, road access is positively associated with the likelihood of farmers purchasing feed by 4.7 percentage points.

Several factors can explain the decline in the reliance on grazing due to road access. For example, improved roads make buying livestock feed easier and convenient. Road access may also increase the opportunity cost of grazing itself, as time spent grazing could instead be dedicated to other income-generating activities enabled by enhanced market access and infrastructure.

Table 3 presents the association between road access and cattle health. Access to roads is associated with an 8.4 percentage point increase in vaccination rates. Improved road access allows veterinary personnel to reach remote communities more consistently. It also enables farmers to bring cattle to vaccination points more easily.

Additionally, our analysis reveals that road access is associated with a statistically significant

Table 2: Influence of Road Access on Feeding Practices

VARIABLES	Share of Grazing in Total Feed	Feed Purchase (1 = Yes)
Road	-0.057*** (0.008)	0.047*** (0.011)
Controls	Yes	Yes
Observations	128,497	128,497
<i>First-stage regression summary</i>		
Coefficient on instrument		0.331***
Standard error		(0.004)
Partial R-squared		0.492
F-statistic (1, 127,974)		7027.450***

Note: The table presents results examining the impact of road exposure on livestock feeding practices, utilizing proxies such as the share of grazing in total feed and feed purchase. The impacts are estimated using a two-stage instrumental variable regression model. See the note under Table 1 for the list of controls. Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3: Effect of Road on Cattle Mortality and Veterinary Utilization

VARIABLES	Cattle Death	Share of Cattle Vaccinated
Road	-0.058*** (0.011)	0.084*** (0.014)
Controls	Yes	Yes
Observations	128,497	128,497
<i>First-stage regression summary</i>		
Coefficient on instrument		0.331***
Standard error		(0.004)
Partial R-squared		0.492
F-statistic (1, 127,974)		7027.450***

Note: The table presents results examining the impact of road exposure on livestock welfare, utilizing proxies such as mortality and vaccination rates. The impacts are estimated using a two-stage instrumental variable regression model. See the note under Table 1 for the list of controls. Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

reduction in cattle mortality. The probability of experiencing at least one cattle death decreases by 5.3 percentage points. The significant association between the reduction in the reported cattle mortality and improved road access could be due to improved access to veterinary services, better feeding practices, and enhanced market access.

4.2. Robustness tests

In our main analysis, we included both to-be-treated households and those that were never treated as part of the control group. As a robustness check, we restricted the control group solely to households from kebeles that received roads during the study period, excluding those that never benefited from such developments. Given that treated and untreated areas can differ significantly in unobserved characteristics, excluding untreated areas helps mitigate endogeneity issues that could arise from comparing two distinct populations. The results are presented in Tables A6-A8. As shown in the tables, the results remain qualitatively consistent, except in the case of feeding practices.

To further evaluate the robustness of our IV, we reconstructed it in multiple ways. First, we used the least-cost path from the baseline road to each kebele centroid as an IV, disregarding budget constraints and acknowledging that actual government road expenditures may not strictly follow planned budgets. Second, after constructing the least-cost path for each kebele centroid to the baseline road, we calculated road access to rural kebeles based on distance per person along this path to give more to the populated kebeles. This measure utilized population data from the 2007 census. Third, we hypothesized that kebeles near those directly intersected by the baseline road network may have a higher probability of gaining road access due to their proximity. Hence, we grouped kebeles by proximity: the first group included kebeles directly intersected by the baseline road; the second group included kebeles adjacent to the first group; the third group included those adjacent to the second, and so forth. Kebeles were then connected to either the baseline road or new roads, depending on which connection was more cost-effective. This iterative method was repeated for each successive group. The results, as shown in Tables A9 and A17, remain qualitatively identical and quantitatively comparable.

We conducted an additional analysis using kebeles as the unit of observation. For this, we aggregated observations at the kebele level and applied a fixed-effect IV model. Since the CSS used the same EAs while resampling households across survey waves, this aggregation was feasible. In constructing kebele-year level outcomes, we collapse household-level data by taking the mean

of all variables within each kebele and survey year. The results of this kebele-level analysis were consistent with our main findings and are presented in Tables [A18](#) and [A20](#).

5. Conclusion

Studies that examine the effects of rural roads on the agricultural sector focus on crop production, overlooking the livestock sub-sector. We address this gap by investigating the impacts on livestock production in Ethiopia. We find that access to rural roads significantly affects cattle production in the country. More precisely, we find that access to roads is significantly related to the adoption of modern livestock management practices, including the use of better breeds, improved feeding practices, and the utilization of veterinary services. We also find road access to be significantly associated with cattle mortality reduction.

The findings show that rural road development can play a key role in transforming the livestock sector. Given the various contributions of livestock to rural livelihoods, ranging from providing nutritious food to generating income and supporting households in the face of shocks such as drought, the impacts of improved road access can have far-reaching implications for enhancing well-being and reducing poverty in the region.

The study has the following limitations. First, the repeated cross-sectional nature of our data limited our ability to observe household-level changes over time. Second, we also fail to assess the impacts of road access on broader welfare outcomes, such as income growth or improvements in overall household well-being, as the dataset only focuses on livestock variables. Other datasets with additional household measures do not include as many survey waves during the road expansion, making a similar analysis more difficult. Third, our analysis primarily focuses on cattle management due to both data limitations and the nature of livestock production practices in Ethiopia. Third, although the URRAP road expansion program was implemented in phases, the construction activity was uneven across years. During the first one to two years, efforts focused on planning, surveying, and securing materials rather than construction ([Kebede, 2024](#)). Road construction accelerated in the middle years, with most roads completed toward the program’s end. As too few kebeles received roads during the early stages of the program, we failed to utilize a staggered difference-in-differences design. Despite these limitations, our findings provide valuable contributions to the literature on infrastructure development, rural economics, and livestock management.

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A. Tables

Table A1: Household Characteristics Across Survey Years

Year	Female (Head)	Age (Head)	Education (Years)	HH Size	Agricultural-Oriented
2010	0.18 (0.11)	42.43 (4.81)	2.89 (1.43)	5.38 (0.85)	0.84 (0.17)
2011	0.18 (0.11)	42.43 (4.81)	2.89 (1.43)	5.38 (0.85)	0.84 (0.17)
2012	0.20 (0.11)	43.53 (6.95)	3.06 (1.46)	5.31 (0.86)	0.83 (0.18)
2013	0.20 (0.11)	42.74 (4.92)	3.17 (1.65)	5.24 (0.83)	0.84 (0.18)
2014	0.20 (0.11)	43.07 (4.86)	3.64 (4.52)	5.18 (0.83)	0.84 (0.17)
2015	0.20 (0.11)	42.94 (4.99)	3.44 (1.63)	5.18 (0.83)	0.83 (0.17)
2016	0.19 (0.11)	43.60 (4.96)	3.66 (1.71)	5.15 (0.87)	0.82 (0.20)
Total	0.19 (0.11)	42.96 (5.26)	3.25 (2.26)	5.26 (0.85)	0.84 (0.18)

Notes: This table reports mean household characteristics by survey year. Standard deviations are shown in parentheses beneath the means. “Female (Head)” equals 1 if the household head is female. “Age (Head)” and “Education” refer to the age and years of formal schooling of the household head. “Agricultural-Oriented” equals 1 if the household engaged in the production of both crop and livestock.

Table A2: Descriptive Statistics and Mean Comparison by Treatment Status

Variable	Pooled Mean	Control Mean	Treatment Mean	Mean Difference
Cattle size (number of cattle)	5.11	5.38	4.74	0.64***
Share of exotic/hybrid cattle	0.01	0.01	0.01	-0.01***
Share of grazing in feeding	0.60	0.62	0.58	0.04***
Feed purchased (1 = Yes)	0.26	0.25	0.29	-0.05***
Cattle death (1 = Yes)	0.20	0.22	0.18	0.04***
Share of cattle vaccinated	0.33	0.29	0.38	-0.09***
Male household head (1 = Yes)	0.85	0.86	0.84	0.01***
Age of household head (years)	43.55	43.18	44.07	-0.90***
Lagged SPEI	0.17	0.30	-0.01	0.31***
Family size (number)	5.48	5.52	5.43	0.10***

Note: This table reports descriptive statistics for key variables across treatment and control groups. The first column shows pooled means. Columns (2) and (3) report group means for non-treated (0) and treated (1) households, respectively. The last column reports the difference in means, with significance levels denoted as *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All statistics are based on data from 172,846 observations from the annual livestock survey (2011–2016).

Table A3: Annual Road Construction Plan

Region	2011	2012	2013	2014	2015	Total
Tigray	335	515	550	550	550	2,500
Afar	241	371	396	396	396	1,800
Amhara	2,408	3,708	3,963	3,964	3,960	18,003
Oromiya	4,014	6,180	6,606	6,607	6,600	30,007
SNNP	1,873	2,884	3,083	3,083	3,080	14,003
Gambella	27	41	44	44	44	200
Benshangul-Gumuz	241	371	396	396	396	1,800
Somali	401	618	661	661	660	3,001
Dire Dawa	21	33	35	35	35	159
Harar	7	10	11	11	11	50
Annual Total	9,568	14,731	15,745	15,747	15,732	71,523

Note: The table shows the plan set by the Ethiopian Road Authority to be contracted each year, measured in kilometers.

Table A4: Impact of Road Construction Timing on Baseline Livestock Outcomes (Part 1)

Year of Construction	Total Cattle	Share of Exotic/Hybrid	Cattle Mortality
2012	-0.051 (0.469)	-0.192 (0.246)	2.357 (1.606)
2013	-0.193 (0.474)	0.375 (0.758)	0.896 (1.873)
2014	-0.158 (0.609)	0.223 (0.373)	2.334 (2.512)
2015	-0.267 (0.555)	0.344 (0.638)	1.776 (1.901)
2016	-0.018 (0.507)	-0.509 (0.333)	0.103 (2.545)
Observations	866	860	866

Note: The table presents regression results examining the relationship between baseline village-level livestock outcomes and the timing of road construction. Livestock outcomes are measured during the baseline (2010), with the year 2011 serving as the reference category for road construction timing. Variable definitions are consistent with previous tables. Standard errors are shown in parentheses. Statistical significance levels are denoted as: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Impact of Road Construction Timing on Baseline Livestock Outcomes (Part 2)

Year of Construction	Share of Grazing	Feed Purchase	Share Vaccinated
2012	-0.821 (2.873)	-0.386 (2.108)	-1.703 (4.404)
2013	-3.959 (3.398)	-1.195 (2.553)	1.280 (4.714)
2014	-7.527* (4.167)	2.481 (3.620)	-0.721 (7.318)
2015	-2.580 (3.513)	-2.184 (2.917)	-6.323 (5.501)
2016	0.725 (3.593)	-2.306 (2.381)	0.810 (4.546)
Observations	865	866	866

Note: The table presents regression results examining the relationship between baseline village-level livestock outcomes and the timing of road construction. Livestock outcomes are measured during the baseline (2010), with the year 2011 serving as the reference category for road construction timing. Standard errors are shown in parentheses. Statistical significance levels are denoted as: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Effect of Road Access on Livestock Production (Excluding Never Treated Group)

VARIABLES	Total Cattle	Share of Exotic or Hybrids Cattle
Treatment	-0.863** (0.398)	0.039*** (0.008)
Observations	87,864	87,864

Note: The table presents results examining the impact of road exposure on livestock size and type. The impacts are estimated using a two-stage instrumental variable regression model. Kebeles that never received roads during the study period are excluded. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A7: Effect of Road Access on Livestock Feeding Practice (Excluding Never Treated Group)

VARIABLES	Grazing Share	Feed purchased (1 = Yes)
Treatment	-0.095*** (0.016)	0.000 (0.016)
Observations	87,864	87,864

Note: The table presents results examining the impact of road exposure on feeding practice. The impacts are estimated using a two-stage instrumental variable regression model. Kebeles that never received roads during the study period are excluded. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A8: Effect of Road Access on Livestock Welfare (Excluding Never Treated Group)

VARIABLES	Cattle Death (1 =Yes)	Share of Vaccinated Cattle
Treatment	-0.102*** (0.037)	0.251*** (0.041)
Observations	87,864	87,864

Note: The table presents results examining the impact of road exposure on cattle mortality and vaccination rates. The impacts are estimated using a two-stage instrumental variable regression model. Kebeles that never received roads during the study period are excluded. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A9: Impact of Rural Roads on Cattle Size and Type Using Least Cost Path Length Per Person to Allocate Road Access

VARIABLES	Total Cattle	Share of Exotic or Hybrids Cattle
Road	-0.704*** (0.196)	0.018*** (0.003)
Controls	Yes	Yes
Observations	128,497	128,497

Note: This table presents the effects of rural road access on total cattle size and type, using an alternative IV. The IV is constructed using the least cost path length per person to determine the sequence of road allocation to each kebele. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are reported in parentheses. Significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A10: Impact of Road on Grazing Share and Feed Purchase using Least Cost Path Length Per Person to Allocate Road Access

VARIABLES	Grazing Share	Feed purchased (1 = Yes)
Road	-0.057*** (0.006)	0.047*** (0.016)
Controls	Yes	Yes
Observations	128,439	128,497

Note: This table presents results examining the impact of road exposure on livestock feeding practices using an alternative IV. The IV is constructed using the least cost path length per person to determine the sequence of road allocation to each kebele. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are reported in parentheses. Significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A11: Impact of Road Access on Cattle Death and Share Vaccinated Using Least Cost Path Length Per Person to Allocate Road Access

VARIABLES	Cattle Death (1 =Yes)	Share of Vaccinated Cattle
Road	-0.058*** (0.015)	0.084*** (0.016)
Controls	Yes	Yes
Observations	128,497	128,497

Note: This table presents the effects of road access on cattle death and the share of vaccinated cattle, using an alternative IV. The IV is constructed using the least cost path length per person to determine the sequence of road allocation to each kebele. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also incorporates three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are reported in parentheses. Significance levels *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A12: Impact of Road on Cattle Size and Type using the Length of Least Cost Path as an IV

VARIABLES	Total Cattle	Share of Exotic or Hybrids Cattle
Road	-0.760* (0.418)	0.010*** (0.003)
Controls	Yes	Yes
Observations	128,497	128,497
R-squared	0.251	0.157

Note: This table presents the effects of road access on the total cattle count and the share of exotic or hybrid cattle breeds, using the length of the least cost path from the baseline road to kebele centroid as an alternative IV. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A13: Impact of Road on Grazing Share and Feed Purchase using the Length of Least Cost Path as an IV

VARIABLES	Grazing Share	Feed purchased (1 = Yes)
Road	-0.026** (0.012)	0.069*** (0.025)
Controls	Yes	Yes
Observations	128,439	128,497
R-squared	0.210	0.065

Note: This table presents the effects of road access on grazing share and feed purchase, using the length of the least cost path from the baseline road to the kebele centroid as an alternative IV. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A14: Impact of Road on Cattle Death and Vaccinated Cattle using the Length of Least Cost Path as an IV

VARIABLES	Cattle Death (1 =Yes)	Share of Vaccinated Cattle
Road	-0.068*** (0.026)	0.074*** (0.026)
Controls	Yes	Yes
Observations	128,497	128,497
R-squared	0.195	0.195

Note: This table presents the effects of road access on cattle death and the share of vaccinated cattle, using the length of the least cost path from the baseline road to the kebele centroid as an alternative IV. The impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), village-level characteristics (crop suitability for various cash and food crops, and the SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A15: Impact of Road on Cattle Size and Type using Alternative IV construction method

VARIABLES	Total Cattle	Share of Exotic or Hybrids Cattle
Road	-1.296*** (0.228)	0.023*** (0.004)
Controls	Yes	Yes
Observations	128,378	128,378

Note: This table presents the effects of road access on cattle size and type, using an alternative IV. The IV categorizes kebeles into four groups based on proximity to the baseline road network: the first group includes Kebeles on the baseline network, the second includes those adjacent to the first, and so on. For each group, least-cost paths from village centers to the nearest baseline road are computed, connecting neighboring Kebeles to the nearest baseline or newly constructed road. This process repeats for each group. Impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), and village-level characteristics (suitability for various cash and food crops, and SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A16: Impact of Road on Grazing Share and Feed Purchase using Alternative IV Construction method

VARIABLES	Grazing Share	Feed purchased (1 = Yes)
Road	-0.064*** (0.009)	-0.007 (0.022)
Controls	Yes	Yes
Observations	128,320	128,378

Note: This table presents the effects of road access on cattle feeding practice, using an alternative IV. The IV categorizes kebeles into four groups based on proximity to the baseline road network: the first group includes Kebeles on the baseline network, the second includes those adjacent to the first, and so on. For each group, least-cost paths from village centers to the nearest baseline road are computed, connecting neighboring Kebeles to the nearest baseline or newly constructed road. This process repeats for each group. Impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), and village-level characteristics (suitability for various cash and food crops, and SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A17: Impact of Road on Cattle Death and Vaccinated Cattle using Alternative IV construction method

VARIABLES	Cattle Death (1 =Yes)	Share of Vaccinated Cattle
Road	-0.075*** (0.021)	0.114*** (0.023)
Controls	Yes	Yes
Observations	128,378	128,378

Note: This table presents the effects of road access on cattle death and the share of vaccinated cattle, using an alternative IV. The IV categorizes kebeles into four groups based on proximity to the baseline road network: the first group includes Kebeles on the baseline network, the second includes those adjacent to the first, and so on. For each group, least-cost paths from Kebele centers to the nearest baseline road are computed, connecting neighboring Kebeles to the nearest baseline or newly constructed road. This process repeats for each group. Impacts are estimated using a two-stage instrumental variable regression model. Controls include household characteristics (age and gender of the household head, family size), and village-level characteristics (suitability for various cash and food crops, and SPEI). The model also includes three fixed effects: time, district, and remoteness (the interaction between baseline distance and survey year). Robust standard errors are in parentheses. The significance levels *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A18: Impact of Rural Roads on Cattle Size and Type Using Kebele as Unit of Analysis

VARIABLES	Cattle Size	Share of Exotic or Hybrids Cattle
Road	-0.076 (0.189)	0.012*** (0.002)
Controls	Yes	Yes
Observations	8,556	8,466

Note: This table presents the effects of rural road access on livestock size and breed composition, with kebele as the unit of analysis. The estimates are derived using a Fixed Effects instrumental variable regression model. The controls include household characteristics (age and gender of the household head, family size), village-level time-varying factors, and the interaction between baseline distance and survey year. Standard errors are shown in parentheses. *** denotes significance at the 1% level.

Table A19: Effects of Rural Roads on Feeding Habits using Kebele as the unit of analysis

VARIABLES	Feed purchased (1 = Yes)	Grazing share
Road	0.053*** (0.011)	-0.035*** (0.010)
Controls	Yes	Yes
Observations	8,556	8,552

Note: This table presents the effects of rural road access on feeding habits, specifically feed purchasing and grazing percentage, with kebele as the unit of analysis. The estimates are derived using a Fixed Effects instrumental variable regression model. The controls include household characteristics (age and gender of the household head, family size), village-level time-varying factors, and the interaction between baseline distance and survey year. Robust standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table A20: Effects of Rural Roads on Cattle Health using kebele as the unit of analysis

VARIABLES	Share of Vaccinated Cattle	Cattle Death (1 =Yes)
Road	0.075*** (0.016)	-0.015* (0.008)
Controls	Yes	Yes
Observations	8,556	8,556

Note: This table presents the effects of rural roads on cattle health outcomes, specifically the share of vaccinated cattle and cattle death incidence using kebele as a unit of analysis. The estimates are derived using a Fixed Effects instrumental variable regression model. The controls include household characteristics (age and gender of the household head, family size), village-level time-varying factors, and the interaction between baseline distance and survey year. Robust standard errors are in parentheses. *** p<0.01, ** p<0.05, * p<0.1.